

Brandon Lin

Software Engineer

✉ blin1283@gmail.com
🌐 <http://brandonlin.com>

Skills

Programming Languages

C++, Typescript, Rust, Python

Frameworks

WebGPU, WebGL, CUDA, PyTorch, $\text{\LaTeX}$

Interests

Speedcubing, Board Games, Memory Sports, Music

Awards/Achievements

4-time Guinness World Record Holder

for Square-1 Average (Rubik's Cube)

William Lowell Putnam Competition

Top 250 Finisher

YouTube Creator (@BrandonLin1)

9 years with 4000+ subscribers

Instagram Creator (@b_lin_cubing)

3000+ followers

Videos peaking at 18M+ views

Education

University of Pennsylvania

May 2020

M.S.E. Data Science

B.A.S. Computer Science

GPA: 3.99

Minors: Statistics & Mathematics

Senior Thesis: *Differentially Private*

Computation in Graphical and

Networked Domains

Other Experience

World Cube Association

Delegate (2023 - present)

Livestream Commentator (2022 - present)

Competition Organizer (2013 - 2019)

Philadelphia Classic

Co-head Organizer (2018 - 2019)

Professional Experience

Figma, Senior Software Engineer, New York, NY

October 2022 - present

- Tech lead for Figma shaders - WebGPU-powered custom fills and shaders created with Figma AI. Led a v-team of 30+ people in a 3-month period to ship at Figma's annual conference, with a focus on graphics rendering performance, agent performance, and cross-functional agility.
- Working on critical performance optimizations and graphics features in Figma's C++/WASM rendering engine, including threaded canvas rendering of the Figma editor.
- Led technical spike to seed Figma's investments in HTML + canvas hybrid rendering, working with engineers at Google to champion the cutting-edge HTML-in-Canvas Chromium API
- Drove an automated migration of file preview codebase to Typescript via a gradual rollout of a language transpiler to improve developer velocity
- Mentored Figma's first PhD graphics intern to implement cutting-edge graphics research in Figma

Meta (formerly Facebook), Software Engineer, New York, NY

July 2020 - September 2022

- Developed and unified ecosystem for rich data pipeline analysis and authoring through performant C++ SQL compiler, implementing privacy-aware type systems and transformability framework for multiple SQL dialects
- Developed CUDA kernels for new ML feature preprocessing framework, prototype for CUDA-based TorchArrow
- Drove org-wide adoption of PyTorch in large-scale ML models through distributed feature preprocessing frameworks and `torch.fx`

Facebook, Software Engineering Intern, Menlo Park, CA

May 2019 - August 2019

- Develop and design system and UI to monitor ML feature and model lifecycles and their relationships with datasets, feature importances, and feature transformations, used by Ads Ranking and Instagram ML teams
- Created functionality for Facebook's feature compiler to reference user package files in feature transform pipeline generation
- Designed Hack and Python API for computing global feature importances across Facebook's ML ecosystem to allow ML practitioners to understand the impact of their models' features

University of Pennsylvania, Teaching Assistant, Philadelphia, PA

May 2018 - May 2020

- Teaching assistant for a variety of university classes, assisting with homework vetting, grading, teaching weekly recitations and holding office hours
- TA'd in Deep Learning, Introduction to Algorithms, Machine Learning, and Mathematical Foundations of Computer Science

Projects & Research Experience

RetroGame | <https://github.com/esqu1/retro-game>

May 2020

- A NES emulator written in Rust, featuring full graphical support and 6502 CPU ISA emulation

TorchLint | <https://github.com/esqu1/torchlint>

November 2019 - December 2019

- A static analyzer for PyTorch written in Haskell to detect device and dimension mismatches
- Employed advanced functional programming techniques such as applicative functors and state monad transformers to develop parser and linter for PyTorch's intermediate representation

PennOS

October 2019 - December 2019

- Unix-like operating system featuring a multithreaded priority scheduler, user-shell interactivity, job control, FAT-style file system, defragmentation utility, and hierarchical directory structure
- Winner of the best operating system in UPenn's undergraduate OS class (CIS 380 Fall 2019)